SoilGrids access

What is the SoilGrids data-sharing policy?

Since 2019, SoilGrids products are provided under the CC BY 4.0 (publicly accessible environmental data; see also the ISRIC software and data policy). SoilGrids contributes to other public global soil data projects. For a review of global soil mapping initiatives and data sets see: Grunwald et al. 2011, Omuto et al. 2013 and Arrouays et al. 2017.

How can I access SoilGrids?

The latest SoilGrids release can be accessed through the following services:

- WMS: access for visualisation and data overview. Instructions for using WMS with commonly used GIS software can be found here
- WCS: best way to obtain a subset of a map and use SoilGrids as input to other modelling pipelines. Examples on how to access WCS can be found here. In particular, Jupyter notebooks for python and examples with R
- WebDAV: download the complete global map(s) in VRT format. We provide examples to access the data from a file browser and programmatically from from R using terra, R using gdal, python and Linux bash. Each map has three elements:
    - a master VRT file;
    - an OVR file with overviews for faster visualisation;
    - a folder with the GeoTIFF tiles.

  For example, to download the 0.05-quantile prediction of coarse fragments in the 5 cm to 15 cm depth interval the user has to get the files `cfvo_5-15cm_Q05.vrt` and `cfvo_5-15cm_Q05.ovr` and the `cfvo_5-15cm_Q05` folder.

  Each map is about 5 GB. All the maps for a single property, with six standard depth intervals and four quantiles per depth occupies about 120 GB.
- A new web mapping platform, with a download option, is available at SoilGrids.org.
- SoilGrids predictions are been made available on Google Earth Engine as community contributed datasets. Please see here for more details.

Use of the SoilGrids (beta) REST API v2.0

Unfortunately, we are currently experiencing issues with the REST API for SoilGrids, and have decided to temporarily pause the service. We are working on a solution. However, at this time, we cannot provide an estimated timeline for when the service will be restored.

Please note that our API is a complementary service we provide to access SoilGrids. It is still under active development (beta stage). Occasional issues and downtimes may occur. We want to emphasise that it does not come with a guarantee of uptime or other types of support. Please remain informed about its status, especially if you are planning to build a production system based on it.

There are alternative, more stable ways to access SoilGrids that, in this case, may better suit your needs. More information on alternative ways to access SoilGrids can be found above.

We will update as soon as the service is restored.

REST API usage statement

How can I use the Homolosine projection?

To deal with an increasing number of inputs and computation demands, SoilGrids has since 2019 been computed on an equal-area projection. After a thorough comparison, the Homolosine projection was identified as the most efficient in an open source software framework (de Sousa et al. 2019). This projection is fully supported by the PROJ and GDAL libraries; therefore, it can be used with any GIS software.

The Homolosine projection is now included in the PROJ database with the code ESRI:54052.

The actual Spatial Reference System (SRS) of the SoilGrids maps is composed by the Homolosine projection applied to the WGS84 datum. This SRS can be added to the PROJ database (the file named `epsg`) with the following string:

```
# ISRIC - Homolosine <152160> +proj=igh +datum=WGS84 +no_defs +towgs84=0,0,0 <>
```

The verbose Well Known Text (WKT) version of this SRS is:

```
PROJCS["Homolosine",
    GEOGCS["WGS 84",
        DATUM["WGS_1984",
            SPHEROID["WGS 84",6378137,298.257223563,
                AUTHORITY["EPSG","7030"]],
   AUTHORITY["EPSG","6326"]],
        PRIMEM["Greenwich",0,
            AUTHORITY["EPSG","8901"]],
        UNIT["degree",0.0174532925199433,
            AUTHORITY["EPSG","9122"]],
        AUTHORITY["EPSG","4326"]],
    PROJECTION["Interrupted_Goode_Homolosine"],
    UNIT["Meter",1]]
```

The European Petroleum Survey Group (EPSG) never issued a code for this projection. However, some programmes like MapServer require any SRS to be associated with such a code. For that reason a pseudo EPSG code was created to refer to the SoilGrids SRS: `EPSG:152160`.

How can I use SoilGrids in a different projection?

The Homolosine projection is not mandatory in any way. The WMS and WCS publish the SoilGrids maps in the following alternative SRSs:

- EPSG:4326 - the popular Marinus of Tyre projection (aka Plate Carré) applied on the WGS84 datum. It expands the surface area of the globe by 60%.
- EPSG:54009 - Mollweide projection (aka Homolographic) applied to the WGS84 datum (pseudo EPSG code issued by ESRI).
- EPSG:54012 - Eckert IV projection applied to the WGS84 datum (pseudo EPSG code issued by ESRI).

The VRT-mosaics can themselves be easily reprojected using the gdalwarp tool. Considering the size of each mosaic, it is best to require a VRT also as output. For example:

```
gdalwarp -t_srs EPSG:3035 -of VRT ./sand_60-100cm_Q0.5.vrt ./sand_60-100cm_Q0.5_3035.vrt
```

Where is SoilGrids code?

The code used to generate SoilGrids will be released (under GPL3 license) together with the submission of a journal article describing the methodology and the main results.

How can I help improve SoilGrids?

To improve predictions for your country or region, consider contributing/sharing soil profile data to the ISRIC WoSIS database so that your point data can be also used to generate improved predictions. Note that ISRIC will always respect the data policy of the data provider and will not publicly share any data unless written permission is given to us to do so. Agencies that contribute data will be acknowledged and listed as a contributing organisation on the main SoilGrids portal.

How can SoilGrids help me improve soil maps for my country?

For specific regions, SoilGrids predictions can be used as a covariate to improve prediction of soil properties through a digital soil mapping approach. Predictions for an area of interest can be obtained from the SoilGrids WCS and overlaid with point data. The SoilGrids maps can thus be used as covariates to predict or adjust the values of target variables locally, together with locally available covariates not used by SoilGrids. An example of how a regional scale prediction can be conducted is described in Hengl et al. (2015).

The SoilGrids framework is intended to facilitate global soil data initiatives and to serve as a bridge between global and local soil mapping. Contact ISRIC to explore how we may collaborate to generate or update predictions for your area of interest.

Suggested uses of SoilGrids by national and regional agencies include:

- use as a covariate layer for regional mapping;
- filling gaps in country-based (bottom-up) global soil information services such as, for instance, developed by the Global Soil Partnership.

- providing input soil data for nationally determined contributions (NDC's), such as for instance required by the UNFCCC.

Who provided soil profile data for the SoilGrids effort?

SoilGrids draws on a large collection of geo-referenced soil profile data for the world that are managed in WoSIS (World Soil Information Service; Batjes et al. (2024). Populating WoSIS has been made possible thanks to the contributions and shared knowledge of a steadily growing number of data providers; we gratefully acknowledge their contributions.

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